An ASDRP neuroscience portfolio · Amishi Mehta

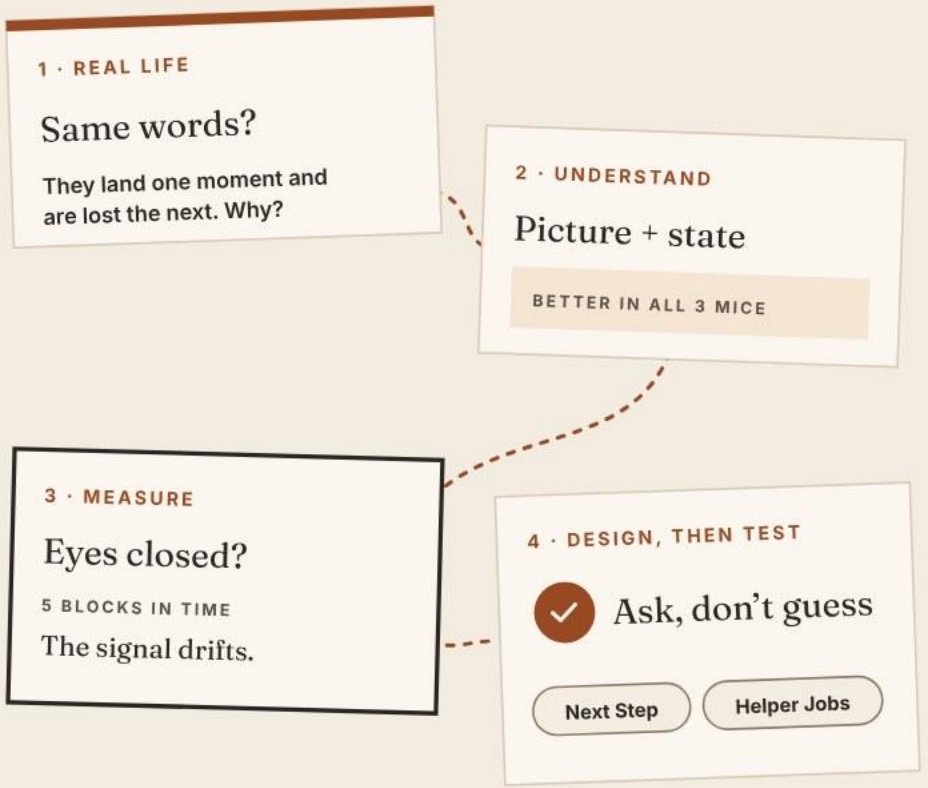
From neurons to next steps

It started from Amishi's own experience: the same instruction can land one moment and be lost the next.

Does what a brain is doing change what it takes in?

And if it does, how should we design for it?

Public data only. A prediction is not a cause. Plans are marked as plans.



In three sentences.

1 · THE MOUSE STUDY

In mice, brain cells reacted differently to the very same picture depending on how alert the mouse was.

Adding how alert each mouse was made our predictions of its brain cells **25–33% better**, in all three mice.

2 · THE BRAIN-WAVE STUDY

A brain-wave headset on a person could not reliably tell even whether their eyes were open or closed.

A computer spotted closed eyes for one stretch, then within two minutes was **no better than a coin flip**.

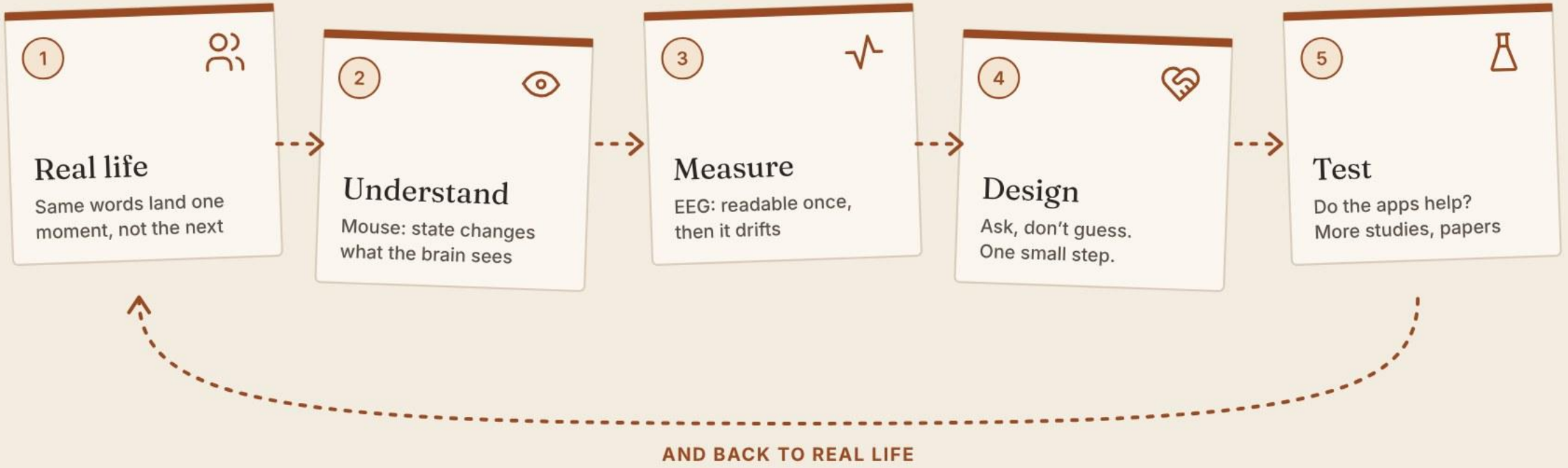
3 · THE APP

So a helpful app should not try to read your brain: it should ask how you feel and show one small step at a time.

Next Step asks **“How much energy do you have today?”** and on a low day shows only the next step.

One loop, five chapters.

It starts in real life and ends back there. Each chapter answers the question the last one raised.



Same words, different moment.



It started from Amishi’s own experience: a step that is easy when you are calm can be lost when you are tired, rushed or overloaded.

The question: is that the brain itself changing what it takes in?
To find out, look inside a brain while its state changes.

Does a mouse's state change how its brain sees?

Mice watched pictures while thousands of brain cells were recorded in the part of the brain that handles seeing.



STEP 1

What the mouse saw

Each picture shrunk to a short summary of its light and dark patterns.

STEP 2

What state it was in

Pupil size (wider usually means more alert) and running speed.

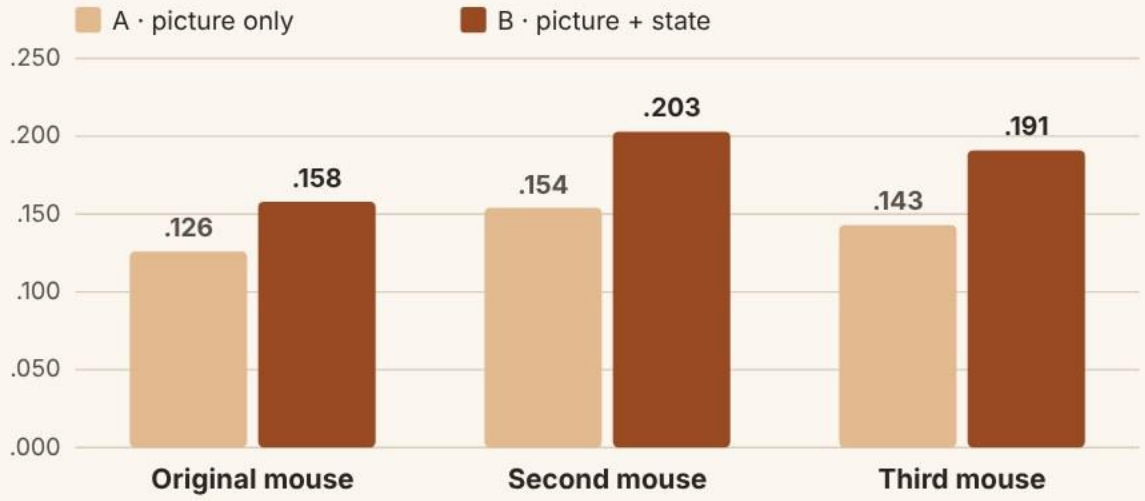
STEP 3

Two simple guessers

A sees only the picture. B sees the picture plus pupil and running.

Every mouse is easier to predict once we know its state.

A quarter to a third better (25–33%), in all three mice, on pictures the model had never seen.
This shows state and seeing are linked. It does not yet show which causes which.



SCORE ON UNSEEN PICTURES

Guesser A against guesser B.

0 means no match, 1 would be perfect. Brain cells are noisy, so scores are low. What matters is the jump from A to B.

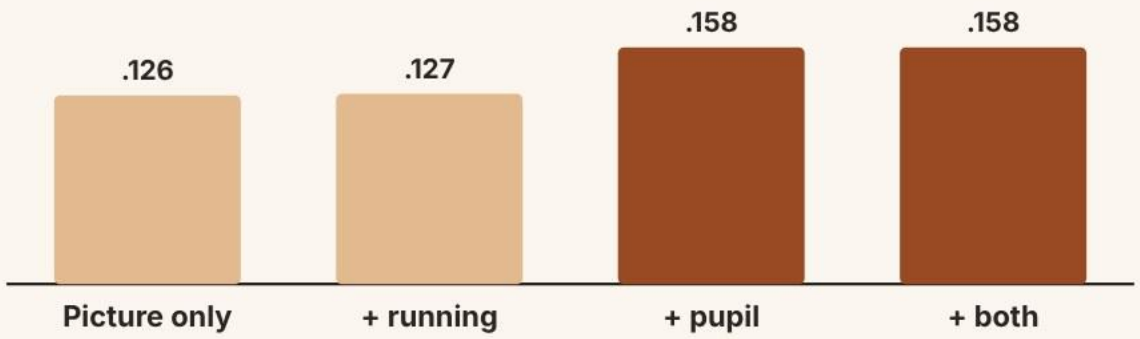
Average over every recorded brain cell.

It is the pupil, not the running.

90%
of brain cells guessed better once state was added

5×
bigger gain for cells that follow pupil size

0.05 → 0.40
one cell, from noise to a pattern



FROM AMISHI'S SAVED RESULTS

Adding pupil gives almost the whole gain.

Running barely moved the score. This mouse ran on only about 5% of trials.

First mouse only, unseen pictures.

Three checks that could have made the gain disappear.

It survived all three.



MIX UP THE STATE

The gain disappears

Pupil and running from the wrong moments add nothing. It needs the state at that moment.

LEAVE ONE MOUSE OUT

Still there

Take any one mouse away and the other two still show the gain.

PUPIL OR RUNNING?

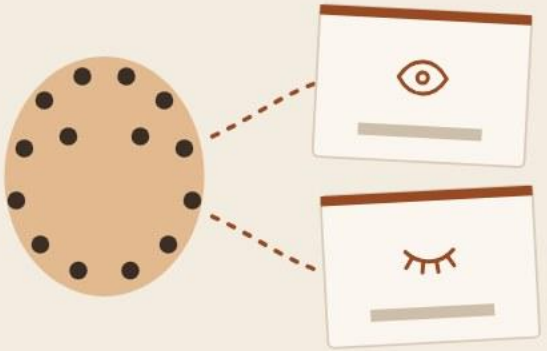
Pupil helps every mouse

Running helps some mice more than others.

What chapter 2 tells us: the same picture lands differently depending on the mouse's state. Can that state be read from outside the head, in a person? Chapter 3 →

Can sensors on the head tell when someone's eyes are closed?

EEG is a cap of small sensors that pick up the brain's electrical activity through the scalp. Closing your eyes is one of the clearest changes it can see.



THE RECORDING

One person, two minutes

14 sensors, about 128 readings a second, each moment marked open or closed.

LEARN, THEN TEST

Earlier, then later

Learn from the first part, test on what comes after, like a real app would.

THE SCORE

0.5 is a coin flip

1.0 is always right. Below 0.5 is worse than guessing.

On one stretch, yes. Over time, no.

On the last part of the recording the model scored **0.625** (0.728 once four broken readings were capped). Across five stretches it averaged **0.478: about a coin flip.**



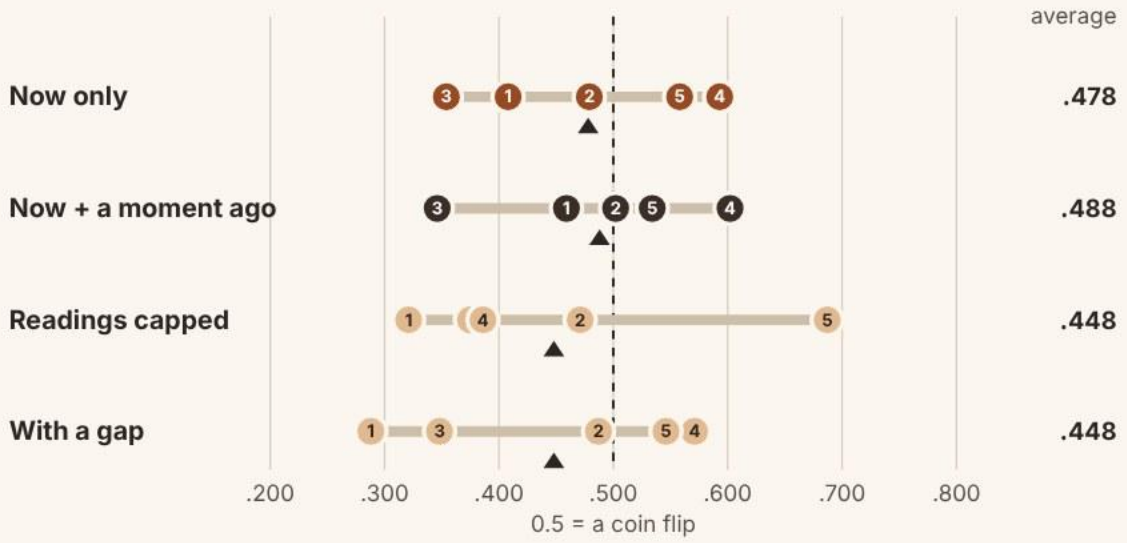
HOW TO READ THE SCORE

Where the results sit between guessing and always right.

Near 0.5 is no better than flipping a coin. Below 0.5 means the pattern it learned has flipped.

The score is called AUC.

Within two minutes, the signal drifts.



FIVE STRETCHES IN TIME

Across the recording, it drifts to a coin flip.

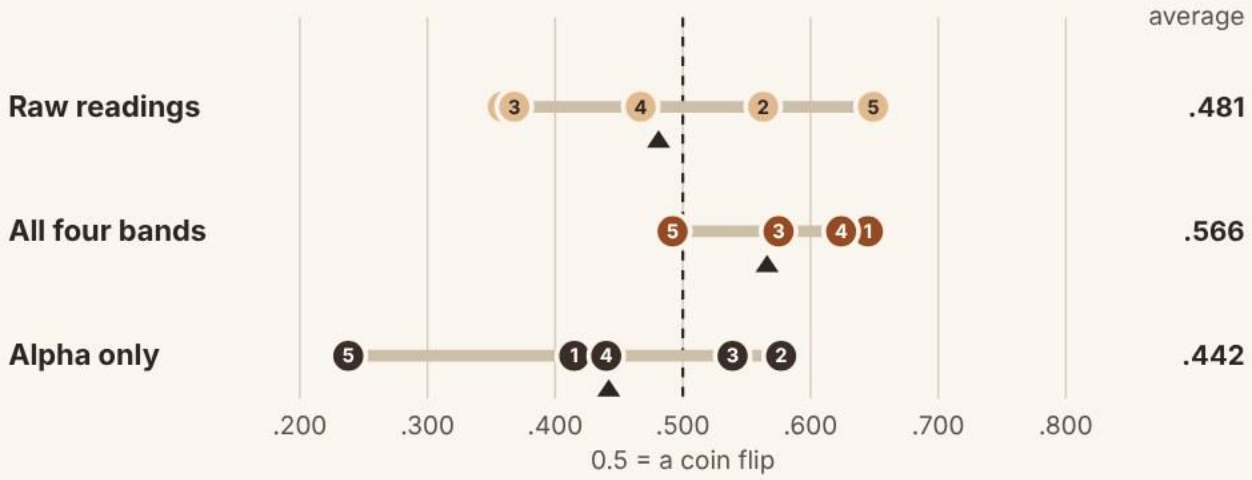
Some stretches fall well below 0.5: the pattern learned earlier had flipped by later on. Capping broken readings and leaving a gap do not fix it.

Mixing up the answers gives 0.499, so the method itself does not invent a pattern.

Would brain-wave bands hold steady?

So the apps ask the person: even bands are not steady enough to build on.

Planned before running. All four bands averaged **0.566** against 0.481 for raw readings, but two of five stretches still fell just below a coin flip. **By the rule set beforehand, it did not hold up.**



FIVE STRETCHES, ONE-SECOND GAP

Better on average, still not reliable.

Alpha alone did worse (0.442). Mixing up the answers gave 0.500, as it should.

Plan and code in the repository.

From findings to features.

FROM CHAPTER 2

Show less when energy is low.

Found: the same picture lands differently depending on state.

In the apps: “How much energy do you have today?” On “A little”, just the next step.

FROM CHAPTER 3

Ask, don’t guess.

Found: a sensor read state once, then drifted, even with brain-wave bands.

In the apps: the person says how they are. Nothing happens until they tap.

FROM CHAPTER 3

Expect change halfway.

Found: within two minutes the pattern shifted.

In the apps: Save & pause, a note of where you stopped, break reminders.

FROM REAL LIFE

When stuck, make it smaller.

Seen: a task too big all at once is where people stop.

In the apps: “I’m stuck” makes the step smaller and keeps your place.

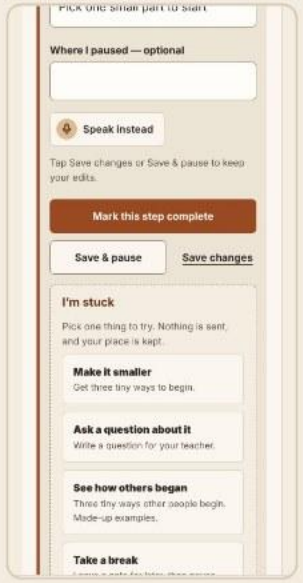
Two apps, built for real life.

Next Step, by Amishi Mehta, for schoolwork. Helper Jobs, a family project, built for real jobs at Pragnya's Sunday outings.



Show less when energy is low

The plan shrinks to one step.



Pause, and make it smaller

Your place is kept.



One picture step, read aloud

With a Help button.

Where this goes next.

NOW · DONE

Two studies, one follow-up, two apps.

The mouse and EEG studies, every result checked and rerunnable, plus experiment 2 on brain-wave bands. Next Step and Helper Jobs working, with the four design rules built in.

Shown: on this site, in the slides and in the phone tour.

NEXT · WITH AN ASDRP MENTOR

Run the three remaining data experiments.

Why the EEG drifts (experiment 4), using the data already here. Then timing (experiment 1, code already written) and groups of brain cells (experiment 3) in the full mouse recordings.

Aim: a first paper, on why the EEG drifts and what brain-wave bands can and cannot fix, for a student research journal such as the *Journal of Emerging Investigators*, plus a poster.

THEN · WITH PEOPLE

Study the design with the people it is for.

Co-design sessions with Pragnya volunteers and families, then a usability test of the four design rules. Consent first, and tasks written down before anyone takes part.

Aim: a second paper, on the design rules, and better apps.

LONGER TERM · THE JOINED-UP QUESTION

Does designing for state help people learn?

Compare the energy-aware design with a fixed one, for the same person over several weeks. Then ask whether the brain findings predict who it helps most.

Aim: one research direction, where neuroscience and human–computer interaction meet.

THANK YOU

From neurons to next steps.

The loop: real life raised the question; the mouse showed that state changes what a brain takes in; the EEG showed that state is hard to read from outside; so the apps ask how someone is doing and keep each step small.
Three mice and one person are a start, not a rule for everyone.

See it all

Website: mehtama1234.github.io/ASDRP-Neuroscience-Portfolio
Phone tour, slides, Next Step and Helper Jobs are all linked from there.

Who did what

The studies were done by Amishi with her father. Software tools, including AI assistants, helped build the site and the apps. Every number comes from saved results.